

Oskar Girardin | Curriculum Vitae

✉ firstname.lastname@inria.fr

Personal Details

Date and place of birth: 07/11/1999 in Munich, Germany

Nationality: German, French, Swiss

Education

INRIA & École polytechnique

PhD in Optimization

Palaiseau, France

May 2024 - present

- Topic: Learning to Optimize: Beyond gradient-based optimization
- Keywords: Evolutionary strategies, Machine Learning, Reinforcement Learning
- Thesis supervisors: Nikolaus Hansen & Dima Brockhoff
- Teaching:
 - Introduction to Probability & Statistics (Fall 2024)
 - Numerical Analysis (Spring 2025)

CentraleSupélec & ESSEC Business School

Master of Science in Data Science & Business Analytics - Grade: 17.94/20

Paris Area, France

Sep 2021 - Dec 2023

- Received the merit-based ESSEC Excellence Scholarship
- Relevant courses: Advanced Deep Learning, Advanced Optimization, Graphical Models, Machine Learning in Network Science

University of Mannheim

Bachelor of Science in Economics - Grade: 1.5 ($\approx 3.5/4.0$)

Mannheim, Germany

Sep 2017 - Jul 2020

- Study focus: Econometrics, Statistics, Behavioral Economics
- Exchange semester at the University of Edinburgh
- Thesis: "Case Based Decision Theory: An application": Development of a theoretical application of a behavioral model, where today's decisions are governed by similar past experiences. (Advisor: Sarah Auster)

European School of Munich

European Baccalaureate - Grade: 88.5/100

Munich, Germany

Sep 2010 - Jul 2017

- Study focus: Mathematics, Physics, Economics

Professional Experience

INRIA

Research internship in black-box optimisation

Palaiseau, France

Sep 2023 - Feb 2024

Study of surrogate modeling in the context of black-box optimization in the RandOpt team. In particular, the CMA-ES algorithm was studied, which is an algorithm inspired by the evolution of species particularly suited for optimizing ill-conditioned and high-dimensional functions. Supervisors: Nikolaus Hansen & Dima Brockhoff

Deutsche Bundesbank

Internship in Banking Supervision & Research

Frankfurt, Germany

Feb 2021 - Jul 2021

Modelling of financial risks using econometric techniques (Fixed Effects estimation, difference-in-differences). Conducted analyses on behalf of the Task Force on Evaluations, which was appointed by the Basel Committee on Banking Supervision, to evaluate the effectiveness of Basel III. Tools: Stata, Excel, R.

UniCredit Bank*Internship in Economic Research*

Timely prediction of GDP ("Nowcasting") using econometric techniques such as OLS and PCA. Maintenance and updating of Macroeconomic Databases. Impactful Data-visualization & coordination of the weekly research publication "Macro & Markets Weekly". Tools: R, Excel, Bloomberg.

Munich, Germany*Oct 2020 - Dec 2020***Private tutor***Mathematics & French lessons for high school students***Mannheim & Munich, Germany***2016 - 2019*

Extra-curricular activities

Cité Internationale Universitaire de Paris (CIUP)*Elected member in the resident's committee*

Elected representative of the residents of the Swedish House in the "Bureau des résidents" (BDR). The BDR represents the interests of 12 000 yearly residents in the board of administration for the National Foundation CIUP. It also organises several cultural, academic and social events on the campus every month.

Paris, France*Sep 2023 - present***Sia Partners***Corporate Research Project (part of the Master's)*

Contributing to the development of a tool to generate GDPR-compliant synthetic datasets using Generative adversarial networks (GAN). Published a series of articles on their data science research platform Heka.ai. Tools: Python

Paris, France*Jan 2023 - Jun 2023***Münchner Tafel***Volunteer worker in a charitable organisation*

The Münchner Tafel is a charitable organisation with the mission to collect qualitative good food, which would otherwise land in the bin, and distributing it to socially and economically disadvantaged people.

Munich, Germany*Jun 2015*

Skills

Languages: French (Mother tongue), German (Mother tongue), English (C1)

Programming languages: Python (Advanced), R (Advanced), SQL (Intermediate), C++ (Beginner), JavaScript (Beginner), Stata, LaTeX

Other Tools: Apache Spark (Intermediate), Pytorch (Intermediate), MongoDB (Intermediate)